

Photosynthesizers

Leaves contain a green pigment called chlorophyll. This traps the energy of sunlight, which is used to build sugars. The process, called photosynthesis, is the origin of virtually all the food chains on Earth.

An indigo flycatcher snatches flies attracted to the foul stench of the Rafflesia flower.

Flies are drawn to the giant Rafflesia flower because it has the odor of their favorite food: rotting meat.

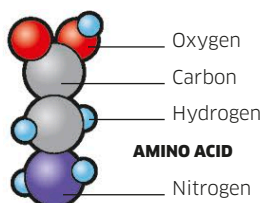
Nutrition

All life needs food—whether it's the sugary sap made in the green leaves of plants, or the solid meals eaten by hungry animals.

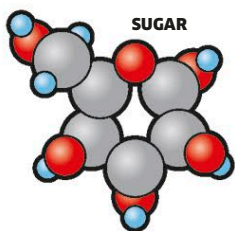
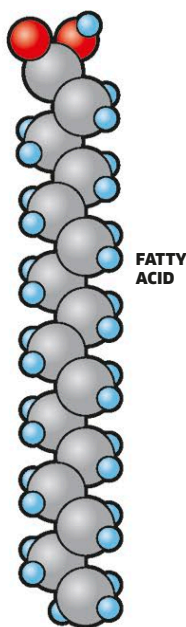
Food gives organisms the fuel to power all the living processes that demand energy, such as growth. Animals, fungi, and many microbes consume it from their surroundings—by eating or absorbing the materials of other organisms, living or dead. In contrast, plants and other microbes start with very simple chemical ingredients, such as carbon dioxide and water, and use these to make food inside their cells.

What is food?

The nutrients in food come from a complex mixture of molecules—each one containing carbon, hydrogen, and oxygen as its main elements. Three main groups—carbohydrates, fats, and proteins—make up the bulk of food molecules, although all organisms require different amounts of each type.

**Proteins**

Groups of atoms called amino acids link into chains of proteins, which help with growth and repair.

**Carbohydrates**

Rings of atoms called sugars provide energy and link to form chains of starch.

Fats and oils

Used for storing energy or building cells, these are made of long molecules called fatty acids.

A Kinabalu pit viper hunts small mammals and birds.

**Mycorrhizae**

A network of fungus filaments—called mycorrhizae—grows among plant roots. Together, roots and filaments have a feeding partnership: the plants pass sugars to the fungi in exchange for minerals gathered by the fungi.

Predators

Animals that prey on others are called predators. Leeches are famous for sucking blood, but the giant red leech has a taste for meat—grabbing giant earthworms as they emerge from burrows after rainfall.